Database ACID Transactions

ACID describes properties that help database transactions remain reliable despite errors and concurrency.

Key facts:

Atomicity means a transaction’s operations are treated as an all-or-nothing unit.

Consistency means a committed transaction moves the database from one valid state to another according to constraints.

Isolation controls how concurrent transactions observe each other’s intermediate or final effects.

Durability means committed data should survive crashes or power loss, usually through logs, checkpoints, or replication.

A transaction groups one or more operations into a logical unit of work.

Indexes can speed up reads but add storage cost and can slow writes because index entries must be maintained.

Normalization reduces avoidable duplication by organizing data into related tables with well-defined keys.

Answerable questions:

Q: What does ACID stand for? A: ACID stands for Atomicity, Consistency, Isolation, and Durability.

Q: Why use indexes? A: Indexes speed up many queries, but they require storage and maintenance during writes.